

Parameter Studies Version 2

Parameter Study – Initial Velocity:

Aim:

To investigate how changing the initial velocity magnitude effects the final y displacement, using a smaller interval than in the previous investigation and letting the curling stone run to the end of its path.

Prediction:

The graph of final y displacement against initial velocity should not be linear and have a one over initial velocity relationship.

Setup:

Parameter	Initial Velocity (m/s)
Minimum	2
Maximum	6
Interval	0.2
Number Of Values	21
Runs Per Value	50
Total Number Of Runs	1050
Other Parameters	Kept Constant

Changes - Increased the maximum runtime so that the path of the curling stone is not cut off

Results:

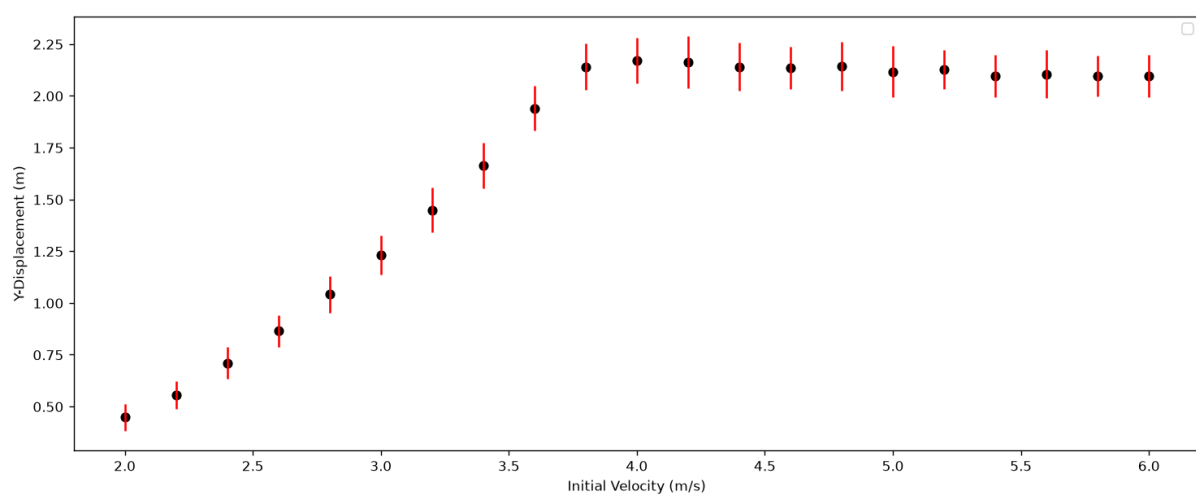


Figure 1: A graph of final y displacement of the curling stone against initial velocity, each point is the average of 50 results, the standard deviation is plotted as error bars.

Total Runtime – 2685s

Conclusion:

The model produces an linearly increasing average final y displacement value between 2 m/s and 3.8 m/s. After 3.8m/s, the final curl value becomes constant then decreases slightly, Therefore, to gain maximum final curl, this model suggests an initial curling stone velocity of between 3.8m/s and 4.5m/s.

Parameter Study – Angular Velocity:

Aim:

To investigate how changing the initial angular velocity of the curling stone effects the final y displacement.

Prediction:

The final y displacement of the curling stone should have a constant value. When the angular velocity is 0, the final y displacement should be zero.

Setup:

Parameter	Initial Angular Velocity (rad/s)
Minimum	0
Maximum	7
Interval	0.5
Number Of Values	15
Runs Per Value	50
Total Number Of Runs	750
Other Parameters	Kept Constant

Changes – the constant A is set to 0 when the angular velocity is 0. This is because there should be no scratches when the stone is not rotating.

Results:

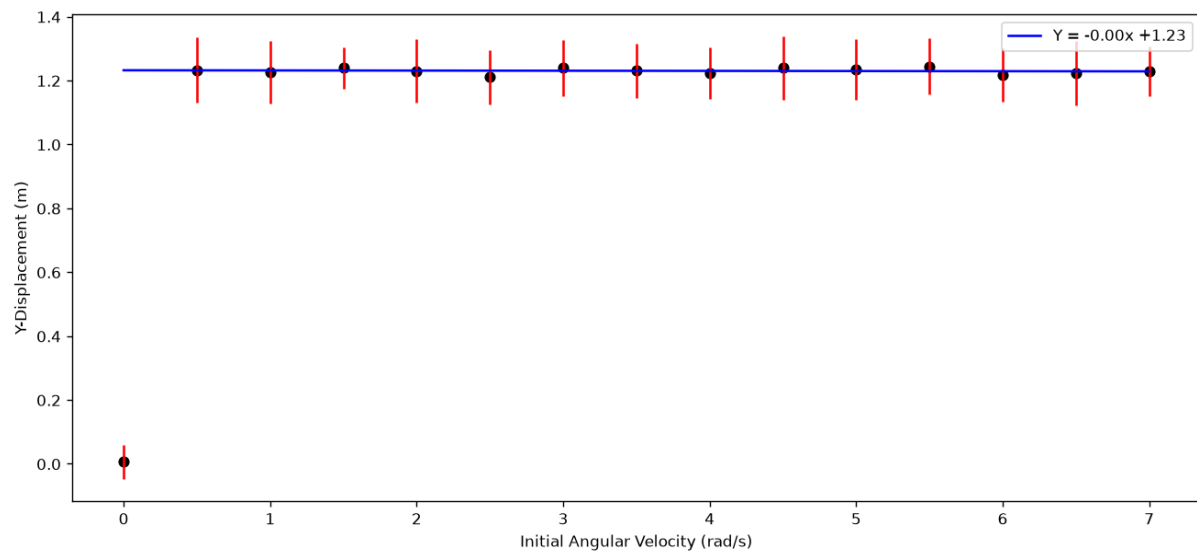


Figure 2: A graph of final y displacement of the curling stone against initial angular velocity, each point is the average of 50 results, the standard deviation is plotted as error bars and a weighted line of best fit is plotted that disregards the first data point.

Total Runtime – 740s

Conclusion:

In conclusion, figure 2 shows that the model produces a constant average curl value for all initial angular velocity values above zero, while produces zero curl when there is no initial angular velocity. This is as predicted. In the future, the value of A should be reliant on the angular velocity therefore removing this straight line relationship.

Parameter Study – Pebble Density:

Aim:

To investigate how changing the pebble density effects the final y displacement of the curling stone.

Prediction:

The final y displacement of the curling stone should have a constant value, as pebble density increases, the standard deviation should decrease.

Setup:

Parameter	Pebble Density Factor
Minimum	1
Maximum	8
Interval	0.5
Number Of Values	15
Runs Per Value	50
Total Number Of Runs	750
Other Parameters	Kept Constant

Changes – Changed how the pebbles touching the back of the curling stone are counted, so that when the total force per pebble is calculated, it is then divided by a different value compared to the previous investigation. This change was applied when investigating initial velocity and initial angular velocity.

Results:

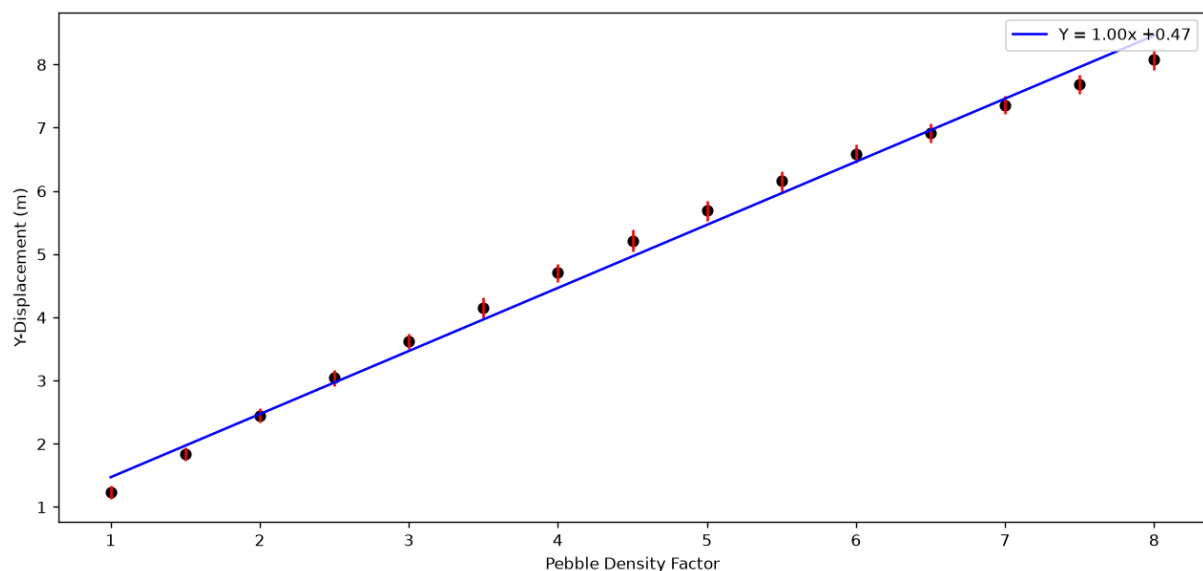


Figure 3: A graph showing the final y displacement of the curling stone against the pebble density factor, each point is the average of 50 results, the standard deviation is plotted as error bars and there is a weighted line of best fit.

Total Runtime – 3712s

Conclusion:

Figure 3 shows that this model produces a linear relationship between the final y displacement and the pebble density. This goes against the predictions as it was expected that the final curl should be a constant value when the pebble density factor was changed.

Overall, the changes made for both angular velocity and initial velocity will be kept while the change made for pebble density will be discarded. All graphs shown in this document were made with all 3 of these changes therefore cannot be compared to the other set of parameter studies.